

STRATEGIC PLANING FOR DATA DRIVEN LOGISTICS OPTIMIZATION

PROJECT 4 – Predictive Modeling and
Optimization in Logistics Systems

PREPARED BY – KHUSHHAL KUMAR SINGH

DATASET USED – DELIVERY LOGISTICS DATA

RECORDS ANALYSED – 25,000

PRIMARY BUSINESS PROBLEM – Can delivery cost be accurately predicted using shipment and logistics characteristics, and how can the resulting predictions be used to optimize logistics operations and reduce costs?

TOOLS USED – Python, Pandas, Pycharm,
Matplotlib

1. Introduction

The systems of logistics produce huge volumes of data concerning different operations such as delivery, transportation, shipment parameters, route planning, vehicle utilization, and cost.

With the help of predictive modeling, this data can be used for analyzing future outcomes and making more efficient operational decisions.

The current project utilizes predictive modeling methods applying logistics data for predicting delivery cost based on logistics and operational parameters. The current research compares various regression models, evaluating them using suitable measures, and determining the options that influence delivery cost the most.

Apart from making predictions, the project focuses on the practical application of the results of the model.

2. Problem Definition

The cost of delivery is a key element of logistics operations because expenses related to transportation and handling have a direct impact on profitability. Delivery cost can vary based on a number of factors including distance, weight of the parcel, mode of delivery used, vehicle utilized, geographical area, atmospheric conditions etc.

This project is aimed at finding a solution for developing a model for predicting delivery costs based on the data available about logistics. The value of predicted cost will allow users to identify their savings opportunities and make logistics operations more effective.

2.1 Business Question

Can delivery cost be accurately predicted using shipment and logistics characteristics, and how can the resulting predictions be used to optimize logistics operations and reduce costs?

3. Project Objectives

- To create a model that can predict the delivery costs.
- To recognize the logistics factors that can affect delivery costs.
- To analyze various regression techniques.
- To assess the performance of the model using MAE, RMSE, and R^2 .
- To perform the cross-validation as well as hyperparameter tuning.
- To distinguish key predictors of delivery costs.
- To suggest useful logistics optimization ideas.
- To recommend useful ideas for the business based on predictive analysis.

4. Dataset Description

The dataset chosen for this project is the Delivery Logistics dataset, which has details about package deliveries, as it is a logistics environment in which factors like various delivery partners, types of packages, vehicles, delivery means, and geographical locations can impact the costs of delivering a package and other delivery results.

The Delivery Logistics dataset has 25000 delivery records and has 15 variables. This dataset is being used to implement a function that will be used for developing a predictive model related to the delivery cost. The relevant shipment and operational variables are taken as predictor variables to derive the required value of the target variable in question, which is the delivery costs.

4.1 Dataset Characteristics

Characteristic	Description
Dataset Name	Delivery Logistics Dataset
Number of Records	25,000
Number of Variables	15
Prediction Type	Regression
Target Variable	Delivery Cost
Primary Objective	Predict delivery cost
Data Domain	Logistics and Transportation

Table1.

5. Variables used in the predictive model

The model used offers a possibility of using variables that can most likely be present at the time of planning or dispatching the shipment. Variables which directly reflect the predicted outcome of delivery are excluded in order to avoid any potential data leaks. The selected variables are comprised of categorical and numerical characteristics of logistics.

Variable	Type	Role
delivery_partner	Categorical	Predictor
package_type	Categorical	Predictor
vehicle_type	Categorical	Predictor
delivery_mode	Categorical	Predictor
region	Categorical	Predictor
weather_condition	Categorical	Predictor
distance_km	Numerical	Predictor
package_weight_kg	Numerical	Predictor
delivery_cost	Numerical	Targe

Table2.

5.1 Target Variable

Delivery Cost (delivery_cost) has been chosen as the response variable for the regression problem. The goal of the study is to predict the expected delivery cost based on the features of a shipment and the parameters of the operation.

5.2 Variables excluded from prediction

Certain variables were left out in terms of prediction since they only serve as identifiers, carry no useful variation, or provide information that can only be available post-delivery. The incorporation of these variables may result in leakages in data, and consequently unrealistic results from the model.

Variable	Reason for Exclusion
delivery_id	Identifier; does not provide meaningful predictive information
delivery_time_hours	No variation in the supplied dataset
expected_time_hours	No variation in the supplied dataset
delayed	Represents a delivery outcome
delivery_status	Represents delivery outcome/status
delivery_rating	Generally available after delivery and unsuitable for dispatch-time prediction

Table3.

6. Data Preparation

The Delivery Logistics Dataset had been analyzed and processed in earlier phase of this project. In order to preserve consistency throughout the project stages, the cleaned data set was deployed for predictive analysis.

The target variable was created separately from predictor variables. The categorical variables were encoded to be machine-readable utilizing one-hot encoding while the numeric variables were prepared to be applied in regression models. The dataset was split using 80:20 train-test ratio for model performance evaluation on test records that had not been seen before.

6.1 Train-Test Split

The dataset was separated into training set and testing set within the ratio of 80:20. The training data has 20,000 observations and is utilized to train the model while the testing data is composed of 5,000 observations which is used for testing the model performance. A random number of 42 is chosen to maintain reproducibility.

6.2 Feature Types

The predictors are made up of six factor predictors and two numeric predictor variables. Categorical predictors need to be transformed into numerical variables prior to being applied in regression algorithms, whereas numeric variables are simply inserted into the model.

7. Data Preprocessing

Prior to modeling, all input variables went through a range of pre-processing procedures. This included one-hot coding categorical variables which produced numerical representation of each input variable by forming binary identifiers for all of its different categories. Standardization of numerical variables was also performed to achieve comparability during the modeling process.

The statistical transformations were incorporated into a pipeline for proper processing of both training and test data and thus avoiding data leakage.

8. Linear Regression Model

Linear Regression was chosen as the initial predictive model. It determines the relationship between the cost of delivery and the predictor variables through the application of a linear equation. Its main benefit is that it can be easily interpreted, which allows understanding the link between the characteristics and the cost of delivery.

This model is being used as a yardstick for testing more complicated methods of machine learning.

8.1 Model Performance

Linear Regression was tested on a 20% testing sample by employing the following methods: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . MAE shows the mean of the absolute differences between the predicted costs and the actual costs, while RMSE takes into consideration large errors. R^2 shows how much differences in delivery costs can be explained by the model.

Metric	Linear Regression
MAE	1.5665224465540206
RMSE	4.671520803962732
R^2	0.999883912452091

Table4.

9. Decision Tree Regression

Decision Tree Regression is the second chosen prediction model after considering its capability of capturing non-linear associations among logistics variables and delivery costs. In contrast with Linear Regression, Decision Tree makes use of decision rules to organize observations in a set of groups enabling the model to illustrate interactions and threshold effects unreachable using a simplistic linear relationship.

The technique was trained using the same training and testing dataset utilized for Linear Regression, therefore, allowing for fair competition between the two models.

9.1 Model Performance

Decision Tree Regression is another method chosen for the purpose of prediction because of its ability to grasp non-linear relationships that exist between the logistics variables and the costs of delivery. In contrast to Linear regression, Decision Tree relies on the rules of the decision-making process which gives it an advantage in classifying observations in the defined groups and helps this method to show effects of interaction and threshold which cannot be observed using simple linear model.

This method was trained on the same training and testing data set as Linear Regression was used thus ensuring fairness in the competition of the two methods.

Metric	Decision Tree
MAE	10.451351157702119
RMSE	12.98331656891761
R ²	0.9991033140678172

Table5.

9.2 Initial Comparison

Model	MAE	RMSE	R ²
Linear Regression	1.5665224465540206	4.671520803962732	0.999883912452091
Decision Tree	10.451351157702119	12.98331656891761	0.9991033140678172

Table6.

10. Random Forest Regression

The Random Forest Regression model was chosen as the third prediction model in the study as it utilizes multiple decision trees for the final prediction. Each tree is trained on different sets of the data and different features to generate the final prediction.

The Random Forest is able to capture the complex non-linear relationships and interactions in logistics variables. This way, it has been included to find out if its more complex model provides a better way of predicting the delivery cost compared to the Linear Regression and Decision Trees.

11. Model performance comparison

The performance of the three regression techniques was assessed using similar test datasets and the same metrics. In evaluating prediction error, different models relied on the same metrics, namely MAE and RMSE, while the effectiveness of each model was determined in relation to the proportion of variability in delivery charges explained by the model (R^2). A model demonstrating a lower MAE and RMSE and a higher R^2 is superior in terms of predictive abilities. Therefore, the comparison made in this way provides a basis for objective selection of the best performing model.

Model	MAE	RMSE	R^2
Linear Regression	1.5665224465540206	4.671520803962732	0.999883912452091
Decision Tree	10.451351157702119	12.98331656891761	0.9991033140678172
Random Forest	70.61853828508455	89.65418733725697	0.9572427052301103

Table7.

Based on the evaluation metrics, the model with the lowest prediction error and strongest R^2 was selected as the final predictive model for the subsequent analysis.

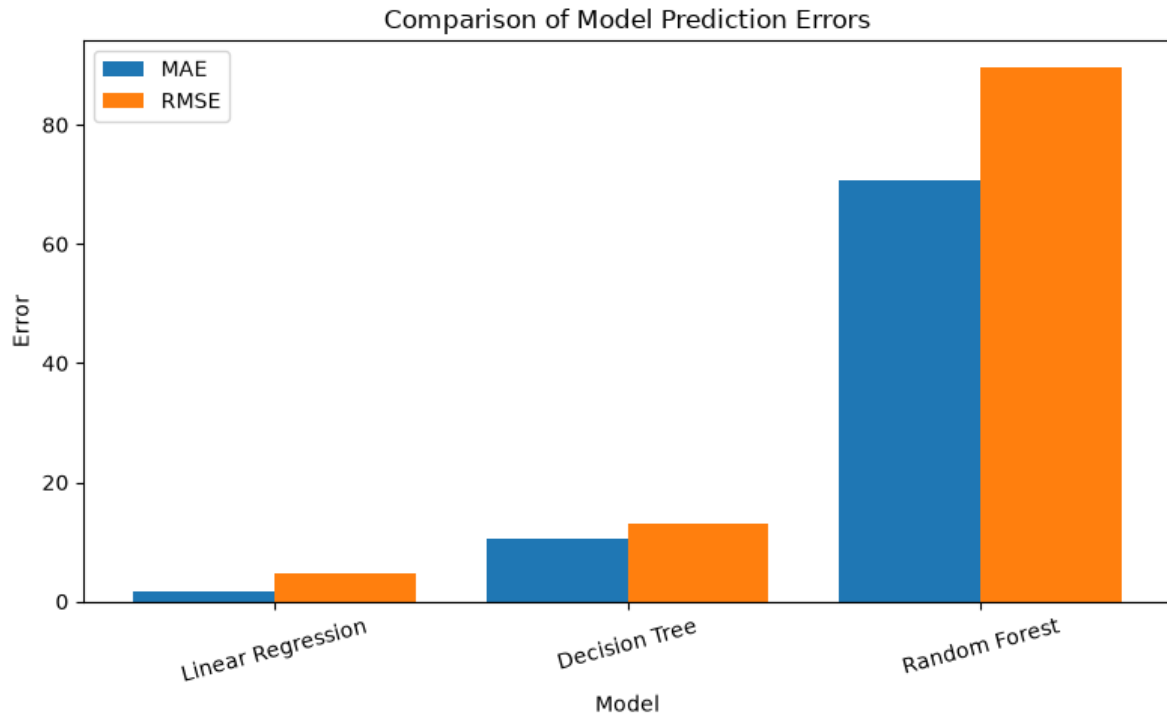


Fig1.

11.1 Interpretation

The comparison of the models reveals how well the three regression methodologies performed in terms of prediction. The more favorable the MAE and RMSE scores, the smaller the predictive errors. The results reveal that having a complex model does not equal having better performance, which suggests that it is necessary to choose the final model based on empirical analysis instead of solely relying on how complex it is.

According to the results of the tests conducted, Linear Regression provided the best performance because of the minimum values of MAE and RMSE indicators and maximum R^2 . The model represented an R^2 of 0.999884 which states that the model provides an explanation of nearly the entire change of delivery costs in the test dataset. Thus, Linear Regression was chosen as the final model for prediction.

12. Cross Validation

Cross-validation means using a method of analyzing the stability and usability of predictive models. The method used was 5-fold cross-validation, where data is split into five parts. The model is trained using four of the parts and tested using one part. The process is repeated until all data sets have been included in the training.

Results from the five training sessions can then be combined to get the average measures of efficiency. This ensures that the outcomes are more accurate than relying on a single train-test combination only.

Model	Mean CV MAE	Mean CV RMSE	Mean CV R ²
Linear Regression	1.557862	5.023811	0.999867
Decision Tree	10.401524	12.904615	0.999122
Random Forest	70.115610	89.166785	0.958103

Table8.

The outcome of the five-fold cross-validation substantiates the effectiveness of the Linear Regression model. The Linear Regression model recorded the least mean value of MAE and RMSE and the maximum mean value of R². The similarity between the validation results and the testing results highlights the reliability of the model.

13. Hyperparameter Testing

The process of hyperparameter tuning was carried out to check the effectiveness of the Decision Tree model by optimizing its definitional parameters. The parameters studied included maximum tree depth and minimum number of samples in a leaf.

The technique called Grid Search along with the validation process was implemented to evaluate various combinations of the variables and determine the configuration that is connected to the least mean value of RMSE.

13.1 Best Hyperparameter

Parameter	Selected Value
Maximum Depth	15
Minimum Samples per Leaf	1

Table9.

Best cross-validation RMSE: **8.906306**

13.2 Tuned Model performance

Metric	Tuned Decision Tree
MAE	5.656094
RMSE	7.397381
R ²	0.999709

Table10.

13.3 Interpretation

Hyperparameter tuning proved to be beneficial for the Decision Tree but did not lead to better performance than Linear Regression. Therefore, Linear Regression is used as a final model for delivery-cost prediction.

14. Final Model Diagnostics

14.1 Actual vs Predicted Delivery Cost

The comparison between the actual and the predicted cost of delivery plot involves an evaluation of predicted delivery costs versus costs observed in the testing dataset. A good model tends to have prediction values that fall close to the diagonal reference line, thus implying that the predicted amounts are close to actual costs.

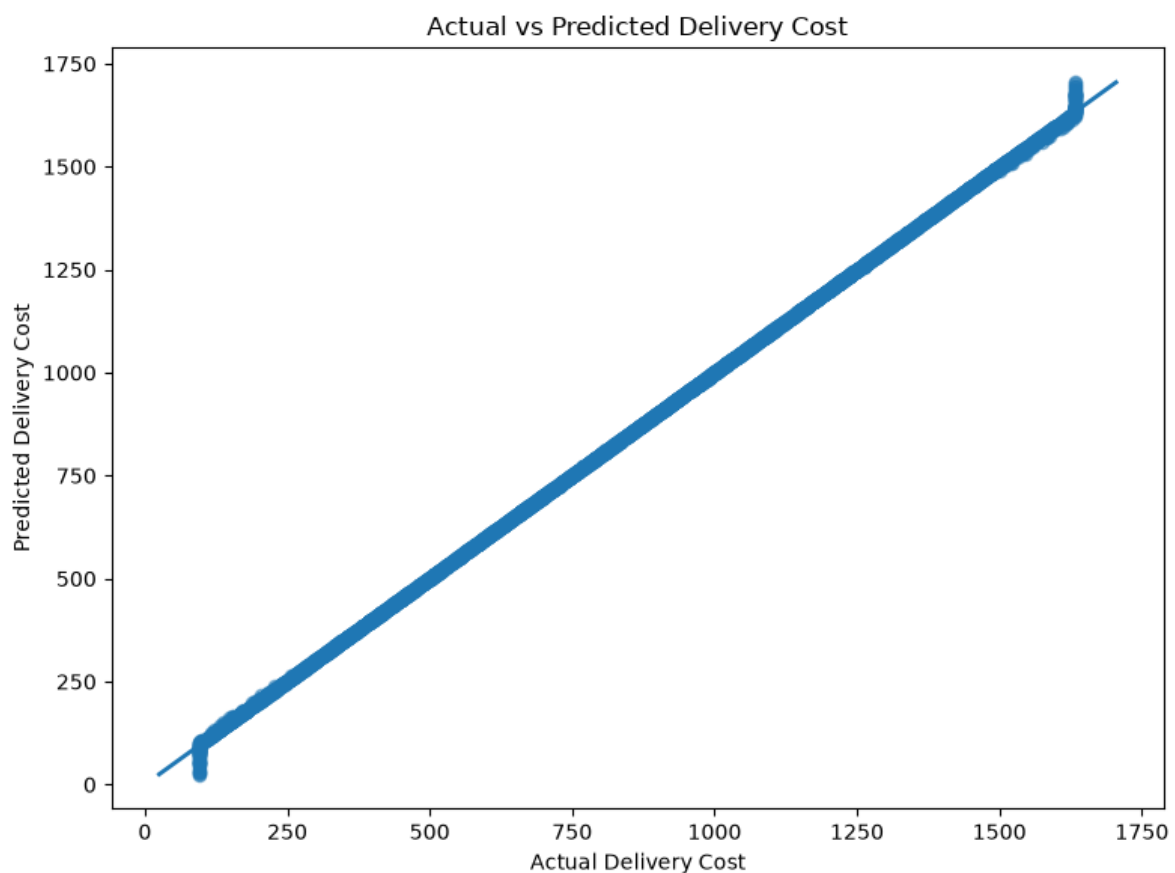


Fig2.

Interpretation

The dataset points are clustered strongly around the diagonal reference line. This shows that the Linear Regression model gives predictions that correlate well with the actual delivery cost amounts. The pattern that can be observed graphically confirms the high R^2 value that we have received from our model.

14.2 Residual Analysis

To investigate the error distribution, residual analysis was conducted. A residual is defined as the difference between the actual delivery cost and the predicted cost. Ideally, residuals should be centred around zero with a weak systematic pattern.

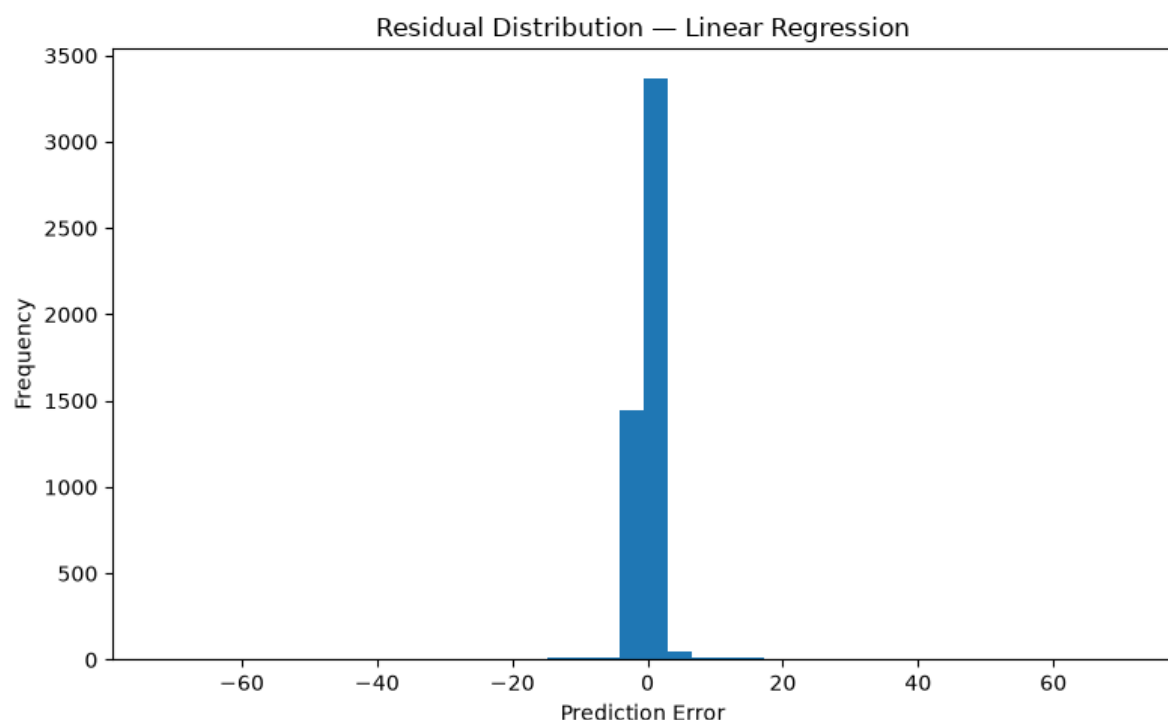


Fig3.

Interpretation

The distribution of the residuals is centered near zero, implying that major systematic errors in the predictions are relatively few. The distribution of the residuals also allows for further verification that the

Linear Regression model is suitable for predicting the delivery cost of this particular data set.

14.3 Interpretation of Predictive Drivers

Analysis of the coefficient reveals that distance and weight are two important quantitative factors of cost. The delivery mode also plays a role in the differences in costs as it implies the additional costs involved in using a service of a different level.

These results allow for establishing a relationship between predictive modeling and logistics optimization. Cost-driving factors may be utilized in the route planning process and in devising the shipment consolidation, in addition to taking into consideration packing methods and delivery modes.

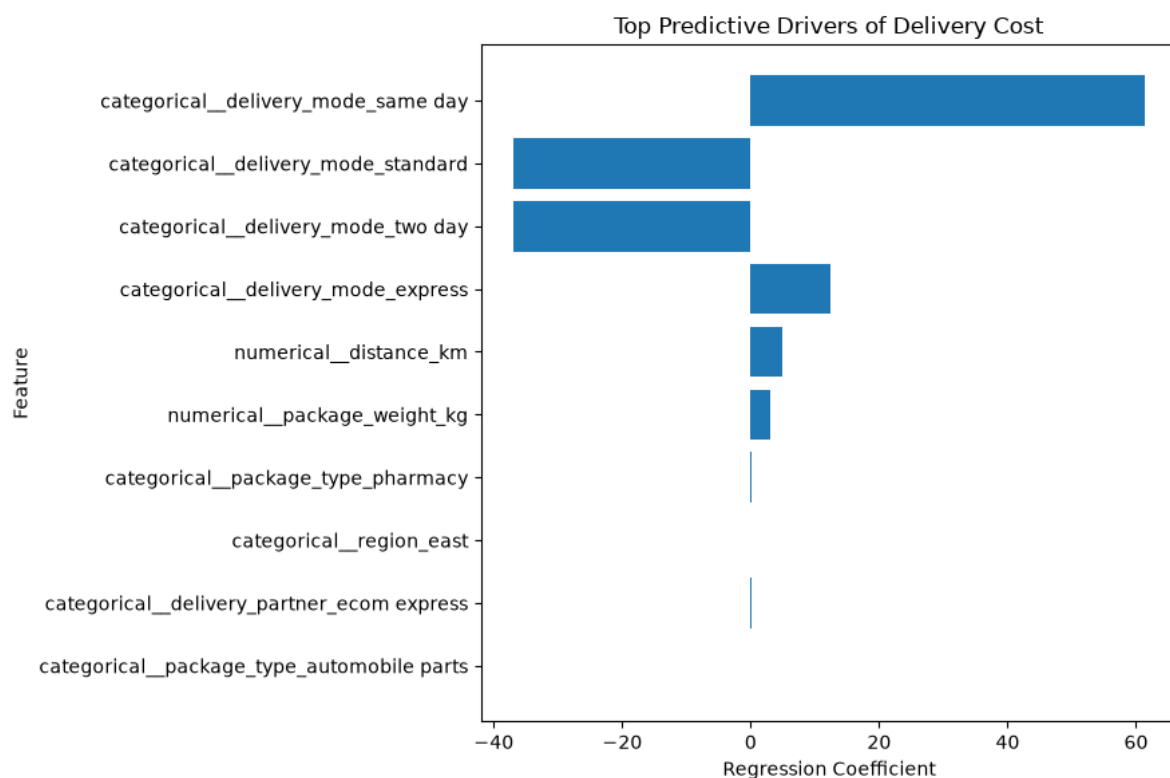


Fig4.

15. Logistics Optimization Strategies

The outcomes generated from predictive modeling may act as a source of information to be used for operational implementation and achieving logistics cost control goal. Given that distance, weight of package and method of delivery were determined to be significant components of the predicted delivery cost, these variables might play a role in formulation of operational decisions and other aspects of operational planning.

15.1 Route and Distance Optimization

Distance can be viewed as one of the major numerical predictors of the delivery cost, meaning that there is a possibility of reducing costs through reduction of unnecessary distance travelled. This can be done through application of route optimization techniques as well as shipment consolidation and using better information for selection of dispatch location.

Before initiating the dispatch of any shipment, it is reasonable to compare various alternative routes on a number of dimensions, such as expected distance and anticipated costs. In addition, shipments with unusually high predicted costs should be flagged for operational review.

Action	Expected Benefit
Optimize delivery routes	Reduce unnecessary travel
Consolidate nearby shipments	Improve vehicle utilization
Use appropriate dispatch locations	Reduce average delivery distance
Review high-distance shipments	Identify expensive deliveries before dispatch

Table11.

15.2 Delivery Mode Selection

Delivery mode is one of the most important variables in delivery-cost estimation. Although the premium delivery mode is one of the fastest services, its operation costs are usually higher. Hence, it is advisable to use the mode of delivery according to the needs of customers and not by default taking the fastest option.

When there is a sufficient level of service provided by means of standard delivery, it may be considered as the least expensive option. Usage of the premium mode of delivery should be justified only in delivery cases where time is crucial or in situations when the clients have specific commitments.

Situation	Recommended Approach
Normal delivery requirement	Prefer standard delivery
Urgent customer requirement	Consider express delivery
Same-day commitment	Use same-day service when necessary
High-cost premium shipment	Review whether premium service is essential

Table12.

15.3 Package Weight and Packaging Optimization

Weight of packages has got a significant impact on the delivery costs involved. It means that there are ways that the entity can adopt to reduce the excessive weight of packages, without compromising the safety of the products being shipped. Innovative packaging, the omission of packaging materials, and consolidation of shipments will help the company control the part of the delivery costs related to the weight.

At the same time, it should be remembered that the weight reduction does not mean putting in danger the products shipped and/or ruining the customer experience. The idea is to eliminate unnecessary weight rather than to reduce package weight.

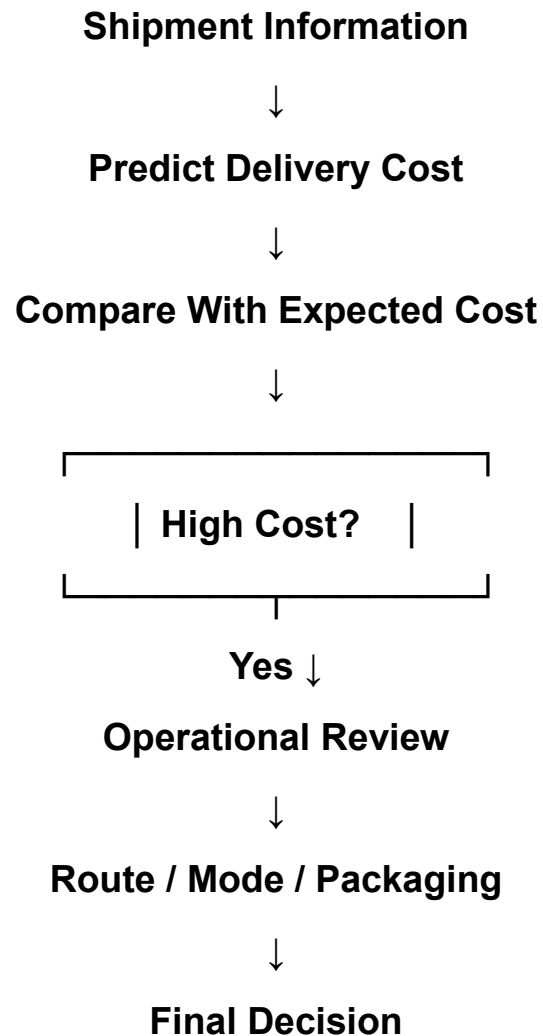
Action	Purpose
Reduce unnecessary packaging material	Lower shipment weight
Review packaging standards	Identify oversized packaging
Consolidate compatible shipments	Improve transportation efficiency
Monitor high-weight shipments	Identify cost-saving opportunities

Table13.

15.4 Predictive Cost Monitoring

The final model of the Linear Regression can be used in the logistics planning process to calculate the expected cost of delivery in advance of sending it off. Unusual shipments should be marked for inspection if they appear to have excessive costs.

This means that a prevention strategy can be developed for logistics managers to be able to consider other routes, transport methods, types of packaging e.t.c.



15.5 Resource Allocation

Estimates of delivery costs using predictive analysis can aid in deciding how to allocate resources. Deliveries can then be classified according to expected costs and operational requirements so that an appropriate combination of vehicles, routes, and modes of delivery can be selected by managers.

Rather than using historical averages as the basis for resource allocation, predictions regarding costs can be a more precise basis for planning deliveries.

16. Optimization Summary

Key Driver	Optimization Approach	Business Objective
Distance	Route optimization and shipment consolidation	Reduce transportation cost
Delivery Mode	Service-level-based mode selection	Avoid unnecessary premium costs
Package Weight	Packaging optimization	Reduce weight-related cost
Predicted Cost	Pre-dispatch cost alerts	Identify expensive shipments
Resources	Predictive allocation	Improve operational efficiency

Table14.

Optimization strategies are closely linked with results of predictive modelling. Rather than implementing general cost-reduction measures, the organization may prioritize operational parameters most strongly influencing transportation costs. Thus, there is an emphasis on a targeted approach to logisticians cost management.

17. Business Recommendations

The results of the predictive modelling generate many usable pieces of advice for the enhancement of logistics cost efficiency. The recommendations found below go in line with the most influential factors set by the linear regression model and take into account the practical needs of business operators.

17.1 Prioritize Distance Reduction

Distance can be considered as the most important quantitative driver of costs discovered by the model and has the index of about 4.99 cost units per kilometre, assuming that other variables monitored in the model do not change. The average distance for deliveries is approximately 150.39 km.

Thus, a theoretical 10% decrease in average distance will lead to the estimated decrease of about 75.06 cost units for one delivery, according to the model. Thus, route optimization has to be viewed as the most important strategy to reduce costs.

Different measures can be undertaken, such as route consolidation, better choice of dispatching locations, and providing fewer unnecessary routes.

Key Insight: A 10% reduction in average delivery distance is associated with an estimated **75.06 cost-unit reduction per delivery** under the model assumptions.

17.2 Optimize Package Weight

Package weight serves as another important numerical predictor with an approximate coefficient of about 2.95 cost units per additional kilo when other predictors remain constant.

It has been established that the average package weight is about 25.15 kgs. In this case, a 10% decrease in average package weight corresponds to an approximate cost-unit drop of around 7.42 units per delivery, under a constant state of other variable factors.

Consequently, it becomes apparent that logistics operations need to look for packaging redesign opportunities or get rid of useless packaging and merge shipments together.

Key Insight: A 10% reduction in average package weight is associated with an estimated 7.42 cost-unit reduction per delivery under the model assumptions.

17.3 Use Predictive Cost Before Dispatch

The last model of linear regression can be used in the logistical planning process to forecast the delivery costs before shipping. This will help to identify potentially costly packages before any resources are allocated.

In the case when there is an expected high cost of shipment, company managers can search for alternate routes, delivery options, arrangements of the delivery materials, or resources usage.

17.4 Apply Service-Level-Based Delivery Modes

It is necessary to choose the method of shipping in accordance with the required customer service level. High level service delivery options should not be selected automatically when a much less costly delivery solution can successfully achieve the level of service required by the customer.

An approach based on service levels can help establish a balance between the service needs of the customers and the cost of shipping. By utilizing estimations of costs and delivery period estimates, the management can check the efficiency of using a more expensive delivery mode.

17.5 Recommended Action Plan

Priority	Recommendation	Reason	Expected Impact
1	Optimize delivery routes	Distance is the strongest numerical cost driver	High
2	Use predictive cost alerts	Identifies expensive shipments before dispatch	High
3	Optimize packaging and weight	Weight contributes to delivery cost	Medium
4	Select delivery modes based on service requirements	Avoid unnecessary premium costs	Medium
5	Improve resource allocation	Match resources to predicted shipment requirements	Medium

Table15.

An effective strategy for the implementation plan involves ranking the interventions in order of expected impact on delivery costs.

Reducing distance must be the top priority in this scenario, as it is likely to generate the highest modelled impact. It is followed by predictive cost monitoring and package weight optimization.

17.6 Interpretation of Quantitative Estimates

The calculated impacts discussed in this section are model-based projections and not actual savings in monetary terms: these estimates describe expected changes in delivery costs based on the given

assumptions and while the influence of the other variables taken into account remains the same and unchanged. Real savings, in actuality, will depend on operational feasibility, customer requirements, and capacities as well as the ability of the organization to implement the changes proposed.

18. Limitations

The predictive model yields remarkable results while implementing a dataset; nevertheless, a few limitations arise in the interpretation of results.

- The analysis is based on the available Delivery Logistics dataset and may not consider all the variables influencing the real-world cost of delivery.
- The model does not account for all operational variables like fuel prices, traffic situation, availability of the driver or congestion on the route on the way.
- Any predictive correlation established by the model is not necessarily linked to a causal relationship.
- 10 percent distance and weight reduction scenarios provide only estimates of sensitivity and explanations of the model will need to be performed in practice.
- The performance of the model may differ in case it is used in future and under different operating conditions.
- Optimisation decisions must include such elements as service quality, capacity, commitments to the customers and operating limitations besides cost.

19. Conclusion

The application of predictive modelling and optimization techniques was illustrated in this study for a logistics cost-management issue. Three regression models were built—Linear Regression, Decision Tree Regression, and Random Forest Regression—and their performance evaluated using MAE, RMSE, and R^2 as metrics.

Out of three models used, Linear Regression proved to be the best performing with an MAE of 1.5665, RMSE of 4.6715, and R^2 of .999884 in testing data. The model's efficacy was also confirmed by five-fold cross-validation.

Delivery distance and package weight have been identified as important numerical variables for the delivery cost. On the basis of this finding, route optimization, packaging optimization, predictive cost-measurement, delivery-mode choice on the basis of service level, and more efficient resource allocation were proposed as potential measures for optimization.

In short, the project shows that predictive analytics can be useful not only in predicting the outcomes in logistics but also in helping with the actual operational decisions led by data.

20. Python Code Appendix

The entire modeling process was carried out using Python, which was helpful in the stages of data manipulation, implementation of the required model, evaluation of the output, cross-validation of the model, tuning of the parameters, and prediction of target outcomes using the model. The coding snippets exhibit the process and steps occurring in coding.

20.1 Feature and Target Selection

```
features = [  
    "delivery_partner",  
    "package_type",  
    "vehicle_type",  
    "delivery_mode",  
    "region",  
    "weather_condition",  
    "distance_km",  
    "package_weight_kg"  
]
```

```
target = "delivery_cost"
```

```
X = df[features]
```

```
y = df[target]
```

20.2 Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    random_state=42  
)
```

20.3 Data Preprocessing

```
from sklearn.compose import ColumnTransformer  
from sklearn.preprocessing import OneHotEncoder, StandardScaler
```

```
preprocessor = ColumnTransformer(  
    transformers=[  
        (  
            "categorical",  
            OneHotEncoder(handle_unknown="ignore"),  
            categorical_features  
        ),  
        (  
            "numerical",  
            StandardScaler(),  
            numerical_features  
        )  
    ]  
)
```

20.4 Linear Regression

```
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
```

```
linear_regression = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("model", LinearRegression())
    ]
)
```

```
linear_regression.fit(X_train, y_train)
```

```
y_pred_lr = linear_regression.predict(X_test)
```

20.5 Decision Tree

```
from sklearn.tree import DecisionTreeRegressor
```

```
decision_tree = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("model", DecisionTreeRegressor(
            random_state=42,
            max_depth=10,
            min_samples_leaf=5
        ))
    ]
)
```

```
        ))
    ]
)

decision_tree.fit(X_train, y_train)

y_pred_dt = decision_tree.predict(X_test)
```

20.6 Random Forest

```
from sklearn.ensemble import RandomForestRegressor
```

```
random_forest = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        ("model", RandomForestRegressor(
            n_estimators=250,
            random_state=42,
            n_jobs=-1,
            max_features="sqrt",
            min_samples_leaf=2
        ))
    ]
)

random_forest.fit(X_train, y_train)
```

```
y_pred_rf = random_forest.predict(X_test)
```

20.7 Model Evaluation

```
from sklearn.metrics import (  
    mean_absolute_error,  
    mean_squared_error,  
    r2_score  
)  
  
mae = mean_absolute_error(y_test, y_pred_lr)  
  
rmse = mean_squared_error(  
    y_test,  
    y_pred_lr  
) ** 0.5  
  
r2 = r2_score(y_test, y_pred_lr)  
  
print("MAE:", mae)  
print("RMSE:", rmse)  
print("R²:", r2)
```

These metrics were calculated for each model to provide a consistent basis for model comparison.

20.8 Five-Fold Cross-Validation

```
from sklearn.model_selection import KFold, cross_validate
```

```
kf = KFold(  
    n_splits=5,  
    shuffle=True,  
    random_state=42  
)
```

```
scores = cross_validate(  
    linear_regression,  
    X,  
    y,  
    cv=kf,  
    scoring={  
        "MAE": "neg_mean_absolute_error",  
        "RMSE": "neg_root_mean_squared_error",  
        "R2": "r2"  
    }  
)
```

20.9 Grid Search

```
from sklearn.model_selection import GridSearchCV
```

```
param_grid = {  
    "model__max_depth": [5, 10, 15, None],  
    "model__min_samples_leaf": [1, 5]
```

```
}
```

```
grid_search = GridSearchCV(  
    decision_tree,  
    param_grid=param_grid,  
    cv=3,  
    scoring="neg_root_mean_squared_error"  
)
```

```
grid_search.fit(X_train, y_train)
```

```
print(grid_search.best_params_)
```

20.10 Scenario Analysis

```
avg_distance = df["distance_km"].mean()
```

```
avg_weight = df["package_weight_kg"].mean()
```

```
distance_reduction = (  
    avg_distance * 0.10 * distance_coef  
)
```

```
weight_reduction = (  
    avg_weight * 0.10 * weight_coef  
)
```

```
print(distance_reduction)
```

```
print(weight_reduction)
```

The scenario analysis estimates the potential change in predicted cost associated with a hypothetical 10% reduction in average delivery distance or package weight.